

Lecture #9

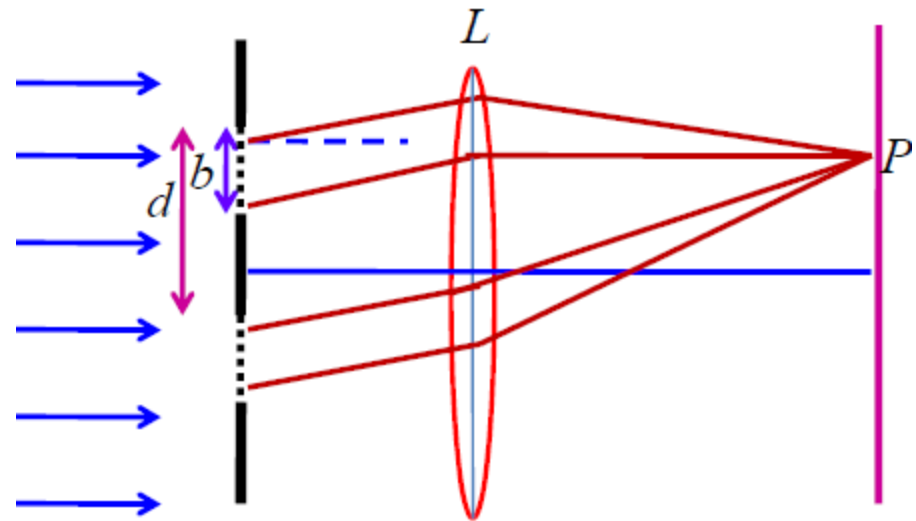
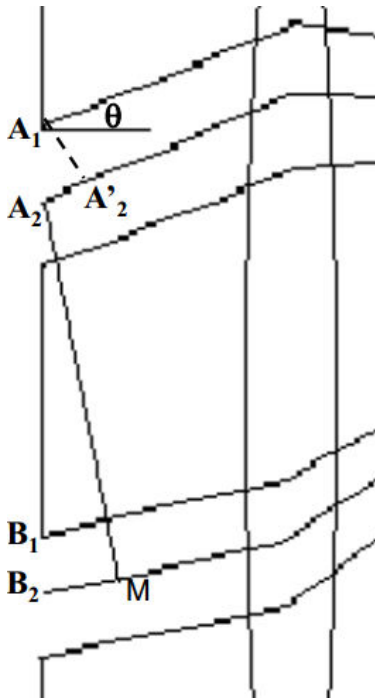
Content to be covered

- ❑ Double slit diffraction
 - ❖ Expression for resultant intensity
 - ❖ Position of maxima (Central and Secondary)
 - ❖ Position of minima
 - ❖ Intensity at secondary maxima
 - ❖ Spread of central diffraction maximum

Resources to be consulted

- ❑ OPTICS by Ajay Ghatak

Fraunhofer Double Slit Diffraction

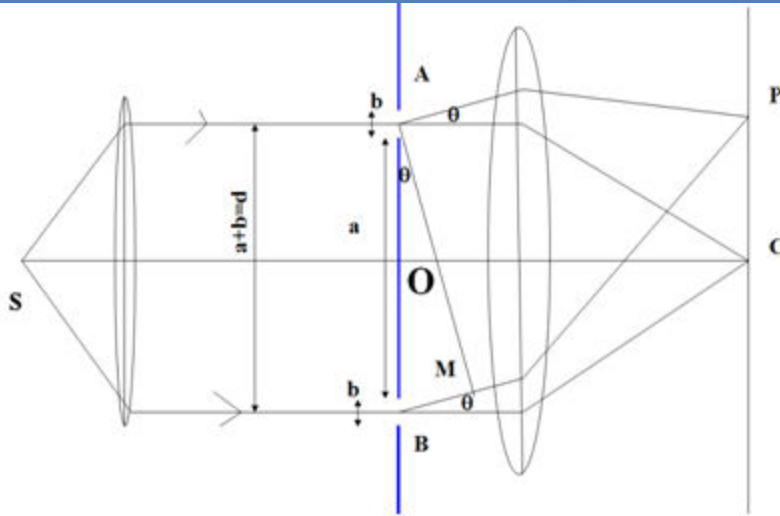


Again, like single slit case, let us assume that slits consist of a large number of equally spaced point sources and each of these points act as a source of secondary wavelets

$$\text{slit width} = 2A_1A_2 = 2B_1B_2 = b$$

$$\text{slit separation } A_1B_1 = A_2B_2 = d$$

Fraunhofer Double Slit Diffraction



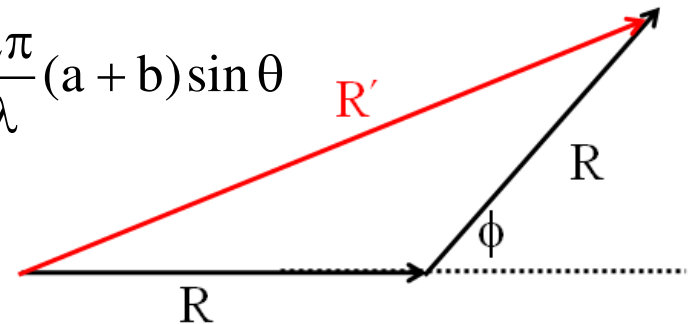
The path difference between the two wave originating from A and B

$$BM = (a + b) \sin \theta$$

$$d = a + b$$

therefore the phase diff. between them is

$$\phi = \frac{2\pi}{\lambda} (a + b) \sin \theta$$



The resultant amplitude due to all the wavelets diffracted from each slit is given by

$$R_P = A \frac{\sin \beta}{\beta}$$

and the resultant phase in this direction is,

$$\beta = \frac{\pi b \sin \theta}{\lambda}$$

Therefore, the resultant intensity at P is

$$I = R'^2 = 4A^2 \frac{\sin^2 \beta}{\beta^2} \cos^2 \frac{\phi}{2}$$

$$R'^2 = 2R^2 + 2R^2 \cos \phi$$

$$= 4R^2 \cos^2 \frac{\phi}{2}$$

$$= 4A^2 \frac{\sin^2 \beta}{\beta^2} \cos^2 \frac{\phi}{2}$$

Fraunhofer Double Slit Diffraction

The intensity distribution will be of the form

$$I = 4A^2 \frac{\sin^2 \beta}{\beta^2} \cos^2 \frac{\phi}{2}$$

Represents diffraction
Pattern produced by
single slit of width b .

Represents interference
pattern produced by
two point sources
separated by distance d .

$$A = na$$
$$\beta = \frac{\pi b \sin \theta}{\lambda}$$
$$\phi = \frac{2\pi d \sin \theta}{\lambda}$$

Position for Minima and Maxima

The expression of resultant intensity for double slit diffraction is given by

$$I = 4A^2 \frac{\sin^2 \beta}{\beta^2} \cos^2 \frac{\phi}{2}$$

Minimum Intensity can be achieved when $I = 0$ i.e.

$$\frac{\sin \beta}{\beta} = 0 \Rightarrow \beta = \pm\pi, \pm2\pi, \pm3\pi, \dots \Rightarrow \boxed{\beta = \pm n\pi}$$

or

$$\cos \frac{\phi}{2} = 0 \Rightarrow \phi = \pm\pi, \pm3\pi, \pm5\pi, \dots \Rightarrow \boxed{\phi = \pm(2m+1)\pi}$$

Now corresponding angular locations of these various **diffraction minima's** would be given by

$$\beta = \frac{\pi b \sin \theta}{\lambda} = \pm n\pi \Rightarrow \boxed{b \sin \theta = \pm n\lambda} \quad \text{where } n = 1, 2, 3, \dots$$

and

$$\phi = \frac{2\pi d \sin \theta}{\lambda} = \pm(2m+1)\pi \Rightarrow \boxed{d \sin \theta = \pm(2m+1)\frac{\lambda}{2}} \quad \text{where } m = 0, 1, 2, 3, \dots$$

Position for Minima and Maxima

Secondary diffraction maxima will occur when

$$\beta = \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \dots = \pm (2n+1) \frac{\pi}{2}$$

$$\Rightarrow \beta = \frac{\pi b \sin \theta}{\lambda} = \pm (2n+1) \frac{\pi}{2}$$

$$\Rightarrow b \sin \theta = \pm (2n+1) \frac{\lambda}{2}$$

where $n = 1, 2, 3, \dots$

Similarly interference maxima will occur when

$$\cos^2 \frac{\phi}{2} = 1$$
$$\Rightarrow \phi = 0, 2\pi, 4\pi, \dots \Rightarrow \phi = \pm 2m\pi$$

$$\Rightarrow \frac{2\pi d \sin \theta}{\lambda} = \pm 2m\pi$$

$$\Rightarrow d \sin \theta = \pm m\lambda$$

where $m = 0, 1, 2, 3, \dots$

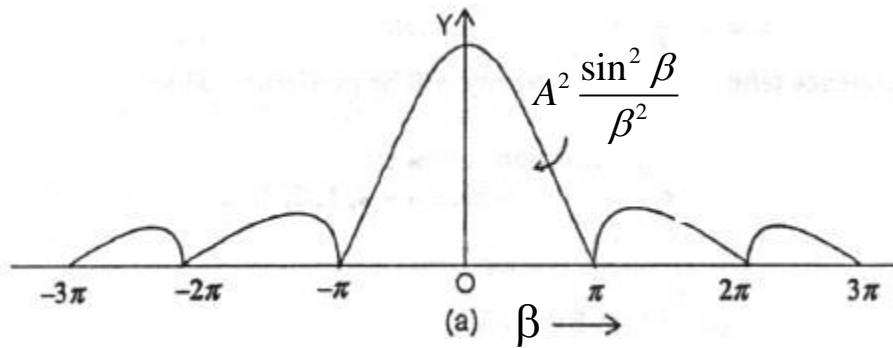
Now again consider interference maxima, we have

$$d \sin \theta = \pm m\lambda \Rightarrow \theta = \pm \sin^{-1} \left(\frac{m\lambda}{d} \right)$$

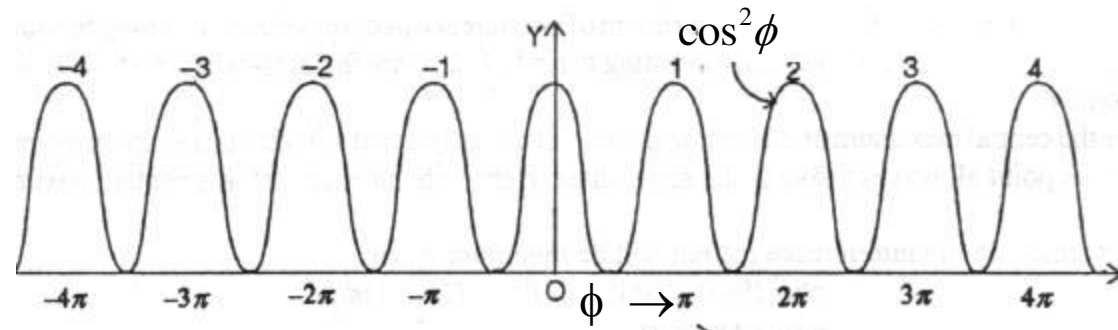
From above expression, we can find out angular position of interference maxima (θ values). Alternatively,

$$\beta = \frac{\pi b \sin \theta}{\lambda} = \frac{\pi b \left(\frac{\pm m\lambda}{d} \right)}{\lambda}$$

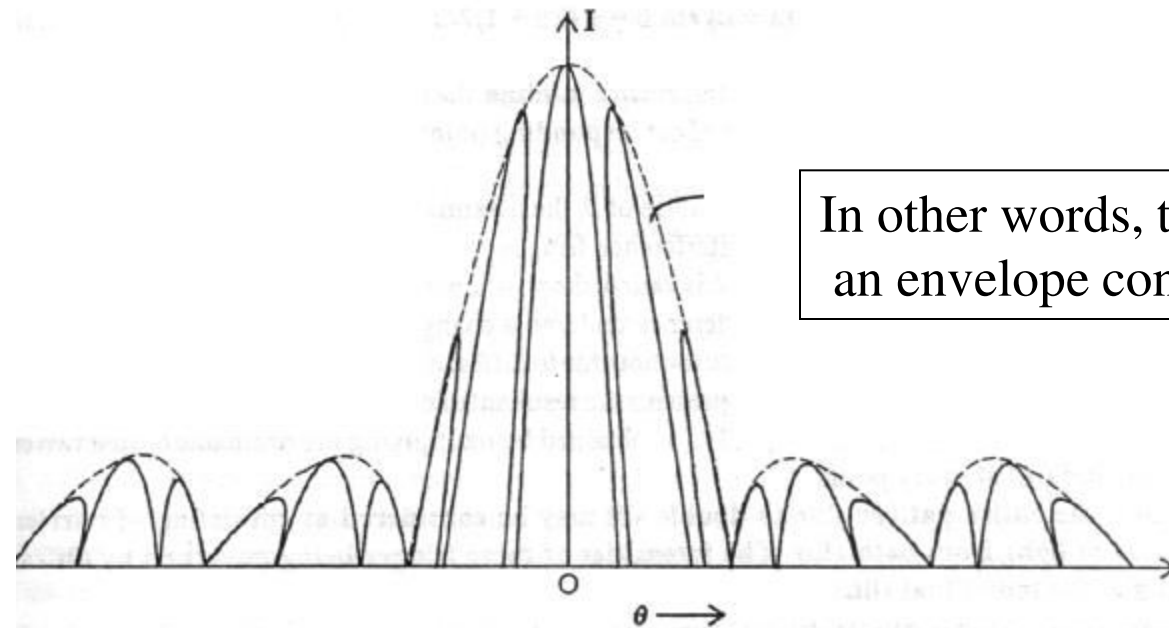
$$\Rightarrow \beta = \pm \frac{m\pi b}{d}$$



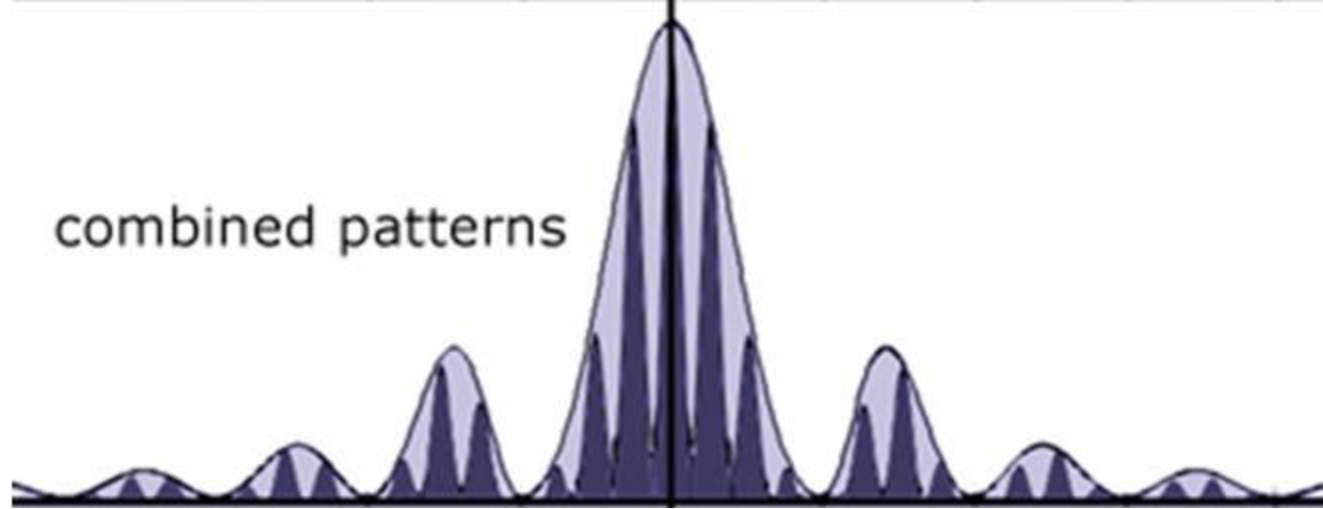
The intensity pattern is in effect a combination of both the single-slit diffraction pattern and the double slit interference pattern.



The amplitude of the diffraction pattern modulates the interference pattern.



In other words, the diffraction pattern acts like an envelope containing the interference pattern.



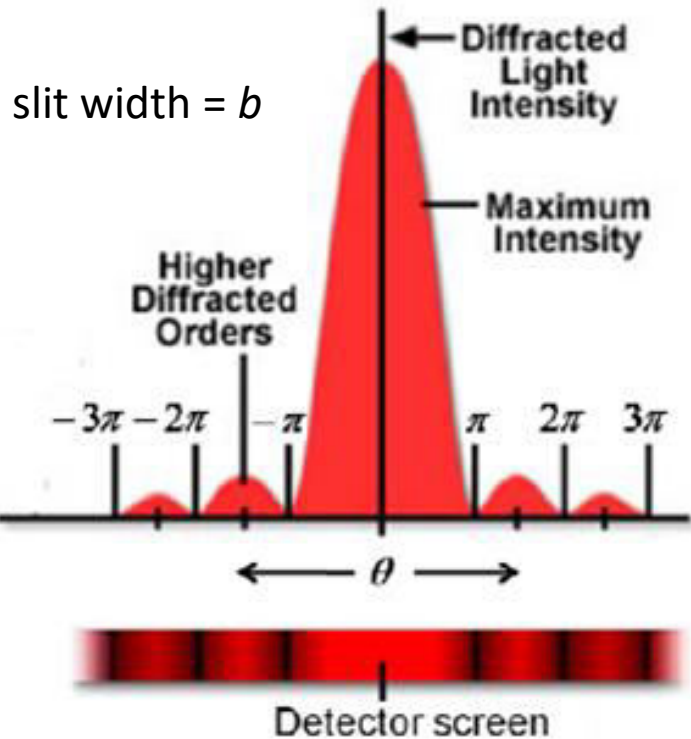
In other words, the diffraction pattern acts like an envelope containing the interference pattern.



The image above is taken from the central maximum area of a display.

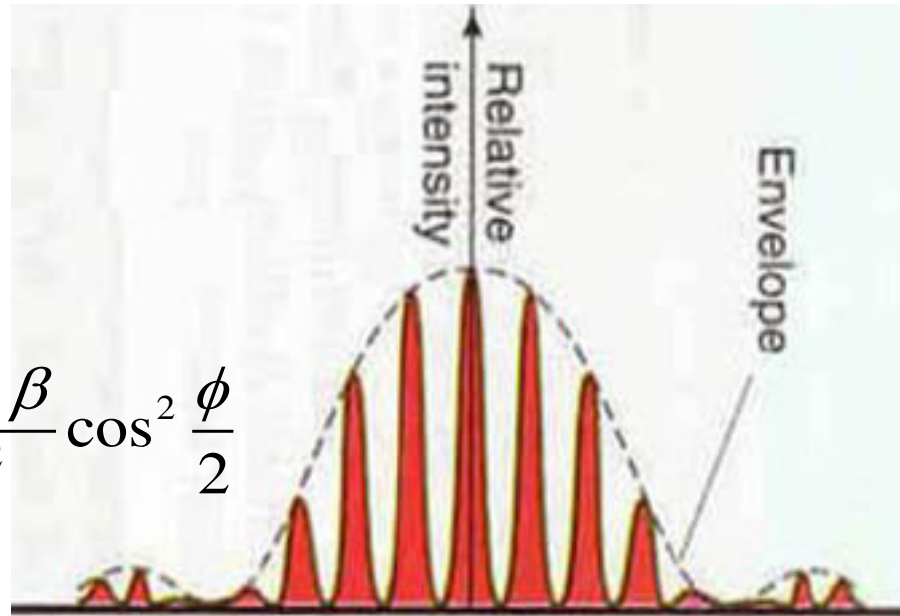
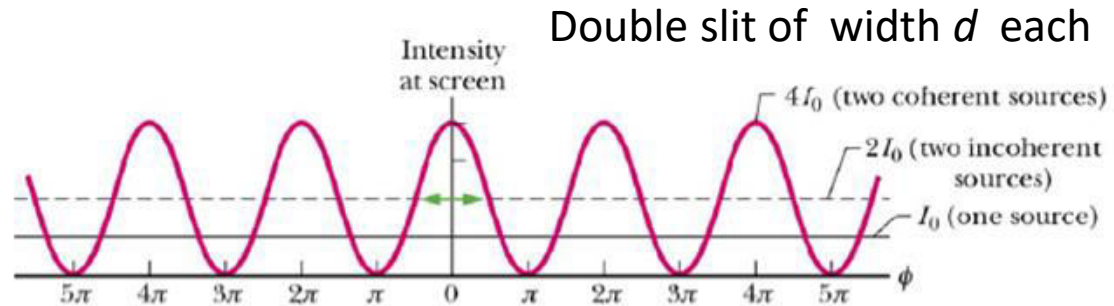
Intensity pattern

Due to single slit diffraction



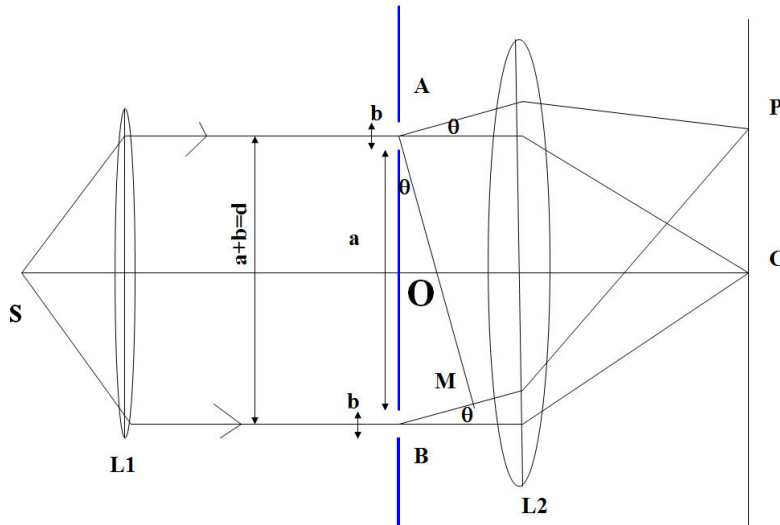
$$I = 4A^2 \frac{\sin^2 \beta}{\beta^2} \cos^2 \frac{\phi}{2}$$

Due to double slit interference



Due to double slit diffraction

FRAUNHOFER DOUBLE SLIT DIFFRACTION



b = slit width
 a = separation b/w slits

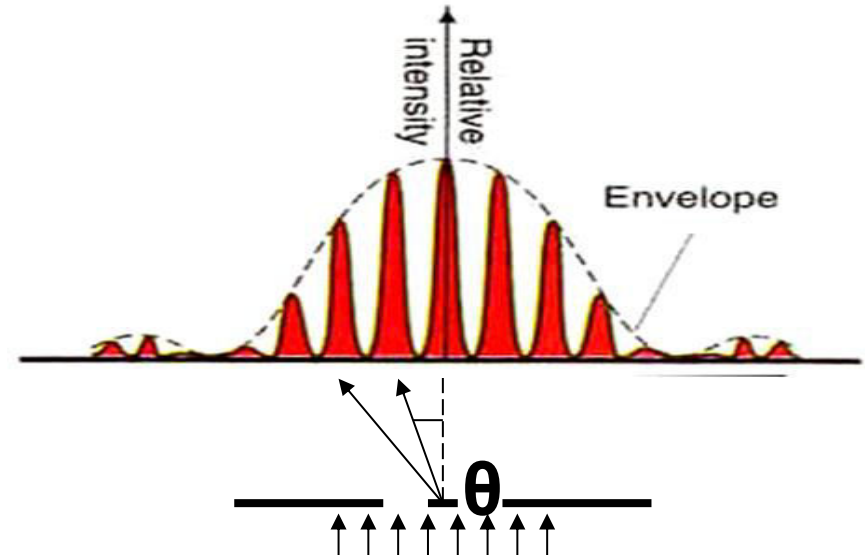
$$d = a + b$$

Diff. Min. $b \sin \theta = \pm n \lambda$

Diff. Max. $b \sin \theta = \pm (2n + 1) \lambda / 2$
 where $n = 1, 2, 3, \dots$

Interf. Min. $d \sin \theta = \pm (2m + 1) \lambda / 2$

Interf. Max. $d \sin \theta = \pm m \lambda$
 where $m = 0, 1, 2, 3, \dots$



$$I = \frac{4A^2 \sin^2 \beta}{\beta^2} \cos^2 \frac{\phi_1}{2}$$

where

$$\beta = \frac{\pi}{\lambda} b \sin \theta$$

and

$$\phi_1 = \frac{2\pi}{\lambda} d \sin \theta$$

Missing Orders In Double Slit Diffraction

Interference maxima positions are given by

$$d \sin \theta = \pm m \lambda \quad \text{.....(1)} \quad \text{where } m = 0, 1, 2, 3, \dots$$

The positions of diffraction minima due to each slit are given by

$$b \sin \theta = \pm n \lambda \quad \text{.....(2)} \quad \text{where } n = 1, 2, 3, \dots$$

Suppose, for a given values of θ , b and d , both equations (1) and (2) are simultaneously satisfied, then we have

$$\frac{d \sin \theta}{b \sin \theta} = \frac{\pm m \lambda}{\pm n \lambda} \Rightarrow \boxed{\frac{d}{b} = \frac{m}{n}}$$

If $d = 2b$, then

$$\frac{d}{b} = \frac{2b}{b} = \frac{m}{n}$$

$$\Rightarrow m = 2n$$

Since $n = 1, 2, 3, \dots$, Therefore, $m = 2, 4, 6, 8, \dots$. Thus, **the even order interference maxima will coincide with 1st, 2nd, 3rd, order diffraction minima due to single slit.** Therefore **three interference maxima will fall within central maxima.**

Missing Orders In Double Slit Diffraction

$d = 2b$		$d = 4b$	
Missing orders of Interference maxima	No. of Fringes falling within central maxima	Missing orders of Interference maxima	No. of Fringes falling within central maxima
2, 4, 6, 8, ...	3	4, 8, 12, 16, ...	7

As the slit separation d increases the number of interference maxima appearing in the region of central diffraction maxima increases.